Problem Solvnig: Search Algorithms

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Outline

1. Introduction
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3. Serach algorithms
4. Uninformed search algorithms
5. Informed search algorithms

Examples of intelligent agents

Examples of intelligent agents.

Intelligent agents are computer programs or systems that can perform actions autonomously, based on their perception of their environment and their ability to reason about that perception. Some examples of intelligent agents include:

1. Chatbots: These are software programs that use natural language processing (NLP) to engage in a conversation with users, simulate human-like interactions and provide assistance.
2. Personal assistants: Voice-activated assistants such as Siri, Google Assistant and Alexa are examples of intelligent agents that can understand and respond to spoken commands.
3. Recommendation systems: These intelligent agents suggest products or services to users based on their past behavior or preferences, such as Netflix's recommendation algorithm.
4. Autonomous vehicles: Self-driving cars use sensors and algorithms to perceive their environment and make decisions about how to drive safely.
5. Trading bots: Automated trading systems that use artificial intelligence and machine learning to analyze data, make predictions, and execute trades in financial markets.
6. Smart home devices: Devices like smart thermostats, lighting systems, and security cameras can adjust to the user's preferences and learn from their behavior to optimize their performance.
7. Game-playing agents: Computer programs that can play games such as chess, Go, or poker at a level that rivals or surpasses human experts.
8. Industrial automation systems: Intelligent agents that can control and optimize the operation of machines and industrial processes, such as robotic assembly lines or smart energy grids.

Types of agents

- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

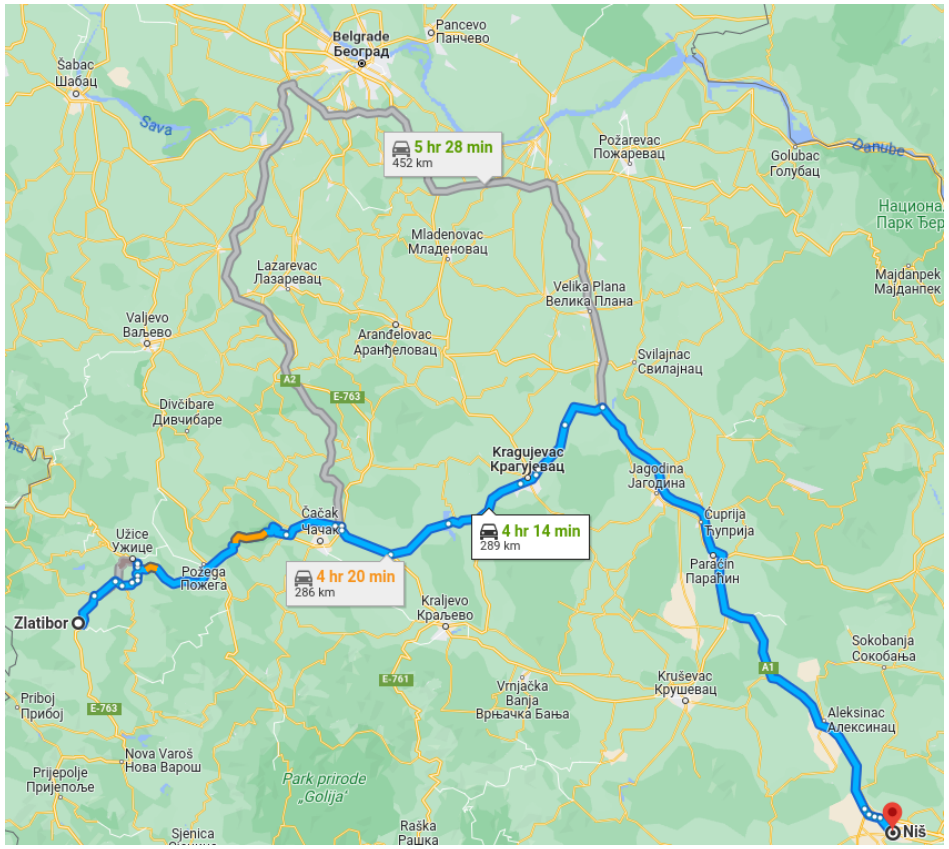
Problem-solving agents

- A special type of goal-based agent
- Goal-based Agents expand the capabilities of the model-based agent by having the "goal" information
- Their choice of action depends on the goal
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not; considerations of different scenario are called searching and planning, which makes an agent proactive

Example: tourist in Serbia

- Tourist is Zlatibor mountin; she can have different goals but she has a flight from Niš tomorrow morning; **Goal formulation** - she needs to be in Niš in time
- **Problem formulation** - given a goal what actions and states to consider; our agent needs to drive to Niš
- If the agent does not have a map the enviroment is **unknown**; our agent has a map - she needs to find a path from Zlatibor to Niš and then carrie out her journey

Example: tourist in Serbia



Agent needs to examine future actions and decided on the best choice of path

- Enviroment is observable, discrete, known, deterministic - the solution to any problem is a fixed sequence of actions
- **Search** - process s of looking for a sequence of actions that reaches the goal
- **Search algorithm** - takes a problem as input and returns **solution**

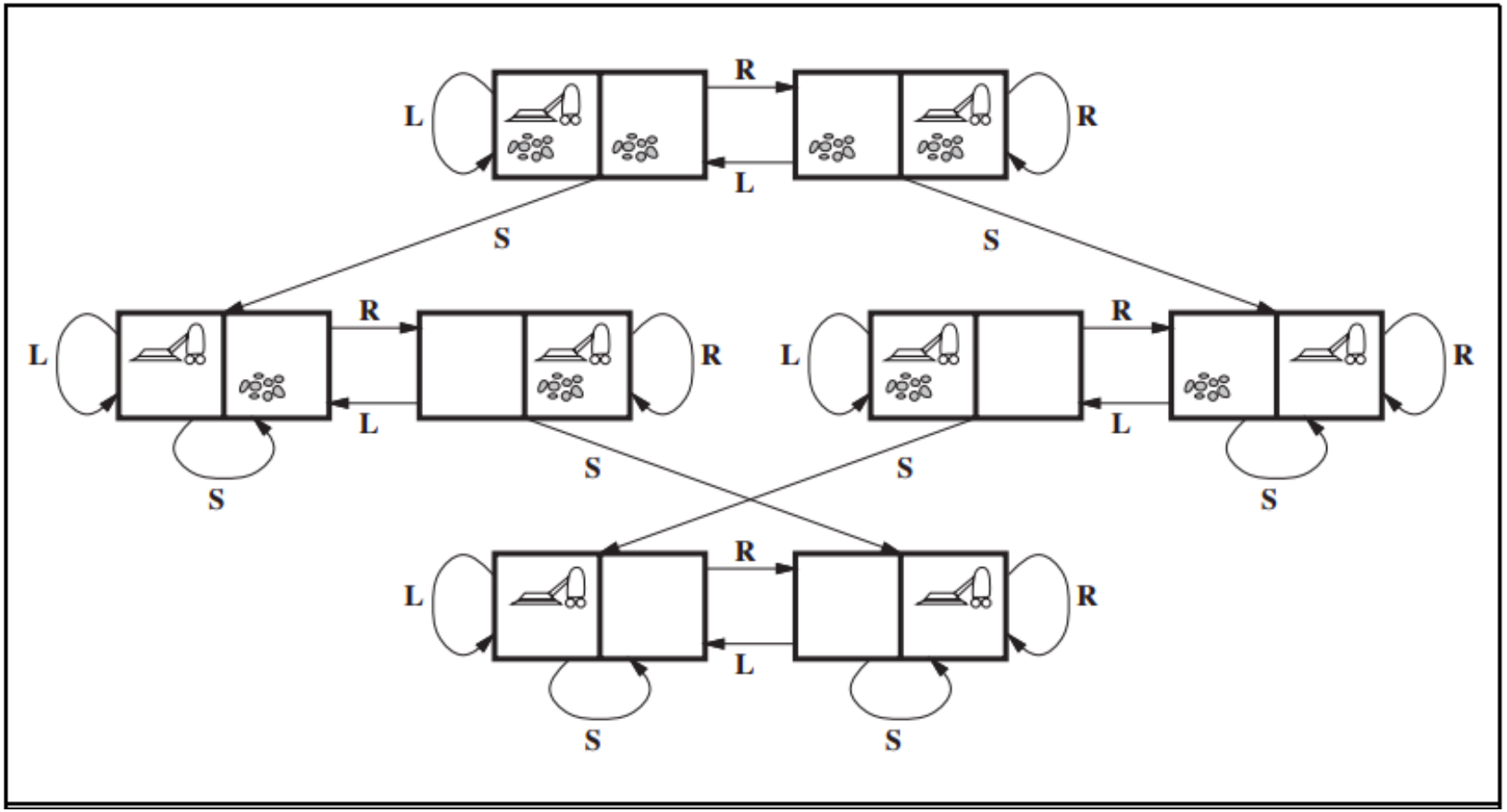
Problem elements

- **Initial state** - the state that agent starts in; Zlatibor for our agent
- List of possible **actions**; s state of the agent - ACTIONS(s) set of actions that can be executed in s;
- **Transition model** - description what action does; **state space** - all states reachable from the initial state; all states create a **graph** - nodes are states and links are actions
- **Goal state** - determines if the given state is a goal state
- **Path cost** - sets a numeric cost of each path

Formulating a problem

- **State abstraction** - simplification of the problem; states can be more complex than the need of the problem
- **Action abstraction** - simplification of actions
- **Level of abstraction** - usage of highroads or driving thourgh mountines; abstraction is useful if carrying out each of the actions in the solution is easier than the original problem;
- **Toy problem** - we omite as much as details as possible but keep the problem and solution; toy problem vs. real-world problems

World vacuum cleaner - state graph



Route-finding problem - airplane route

- States - airport position and time
- Initial state - user's location
- Action - take a flight from current location
- Goal test - is current location goal location
- Path cost - monetary cost, time cost, number of stops, etc.

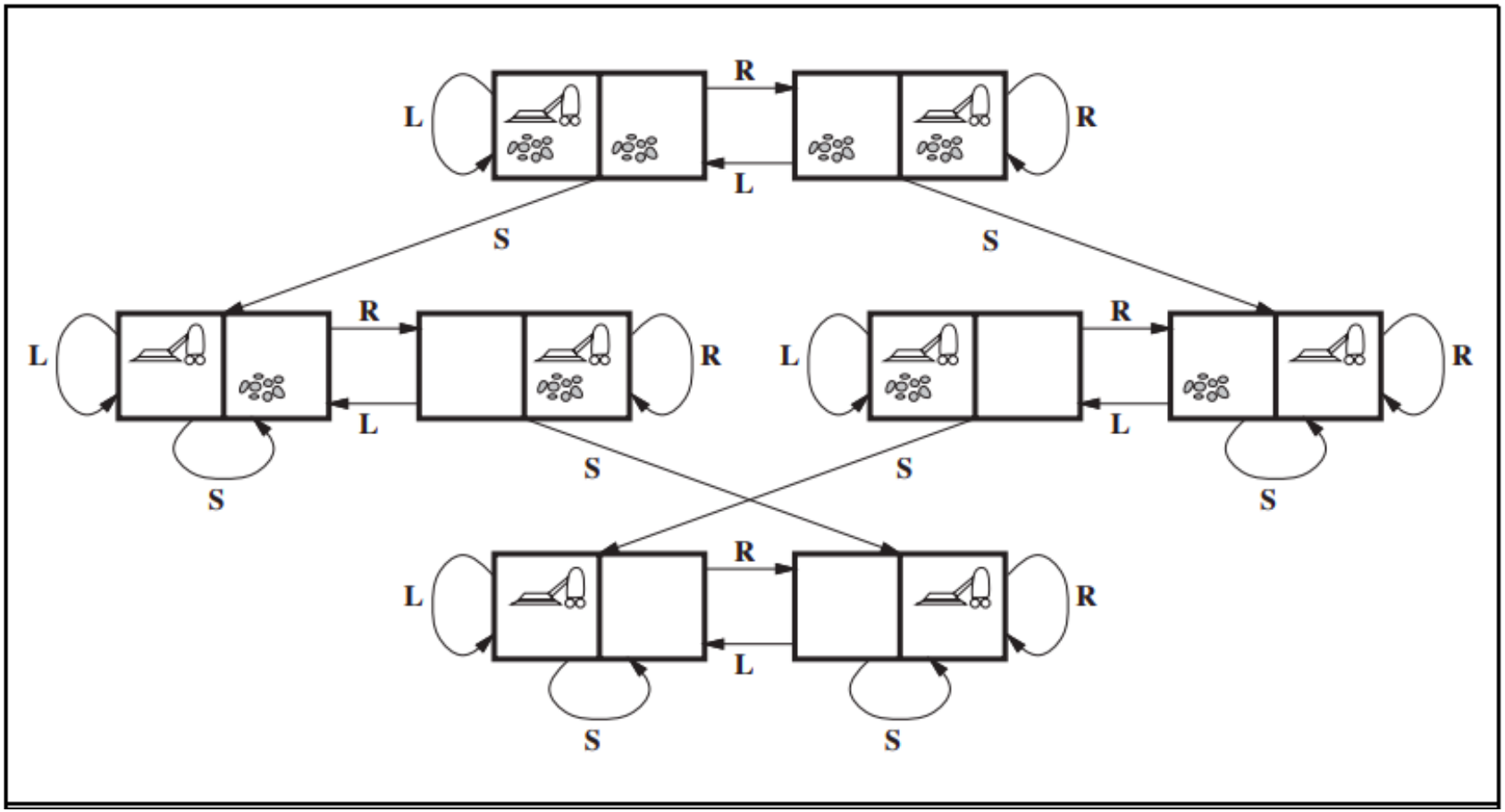
Real-world problems - other examples

- Touring problem - visit every city at least once; we need to known each city that agent visited
- Traveling salesperson problem - touring problem where we visit each city only once
- VLSI layout problem - position of milions components on the limited space; packing problem; cell layout and channel routing
- Robot navigation - route-finding in continiouse space
- Automatic assembly sequencing - findin order in which to assembl an object

Search algorithm - definition

- We have defined problem - soultion is a sequence of actions that solves a problem
- Action sequences form a **search tree** - edges are actions and nodes are states
- First we check if initial state is a goal state; than we **expand** from the initial state; expanding state - apply all legal action to current state and **generate** new set of states; we go from **parent node** to child node
- Essence of search - following one option and put other aside for latter

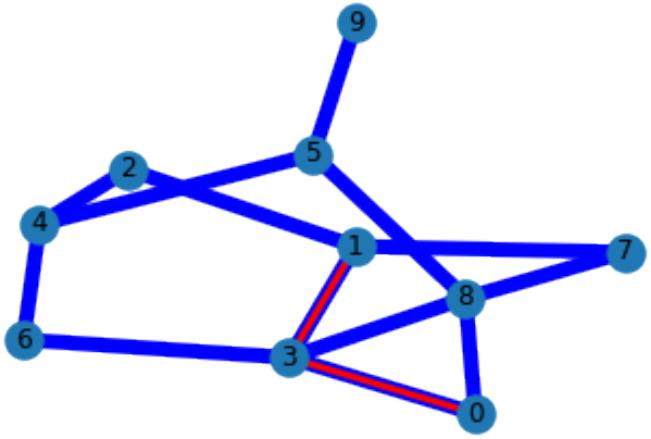
Example: vacuum cleaner

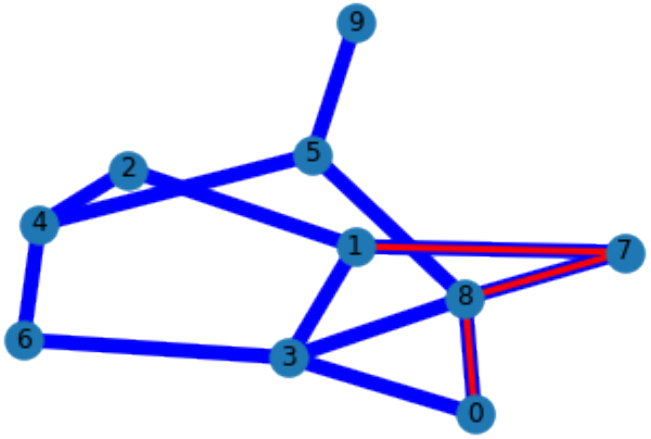


leaf nodes; repeated state; loophy path; redundant path;

Paths in networks

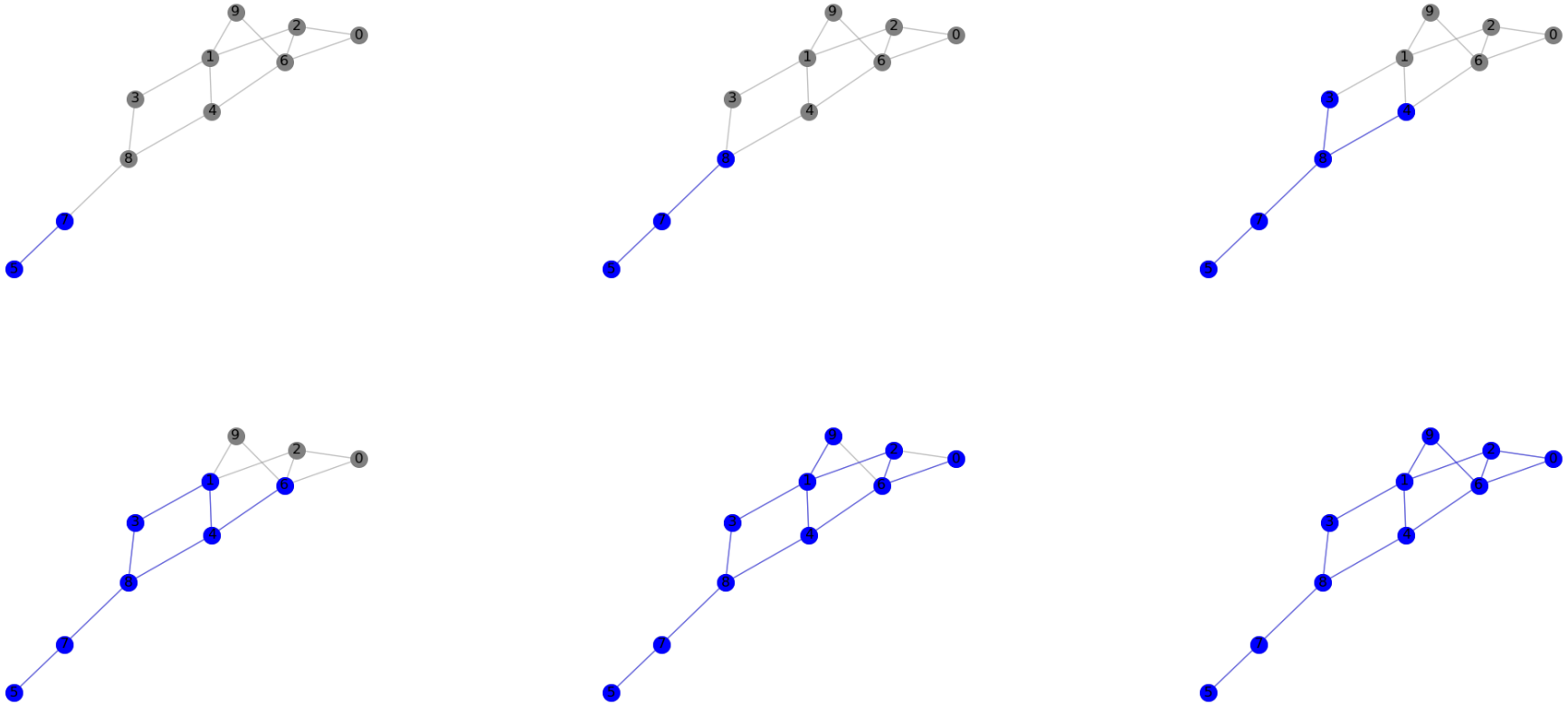
Redundant paths are common





Explored set or **closed list** - list of every expanded state

Graph-search algorithms



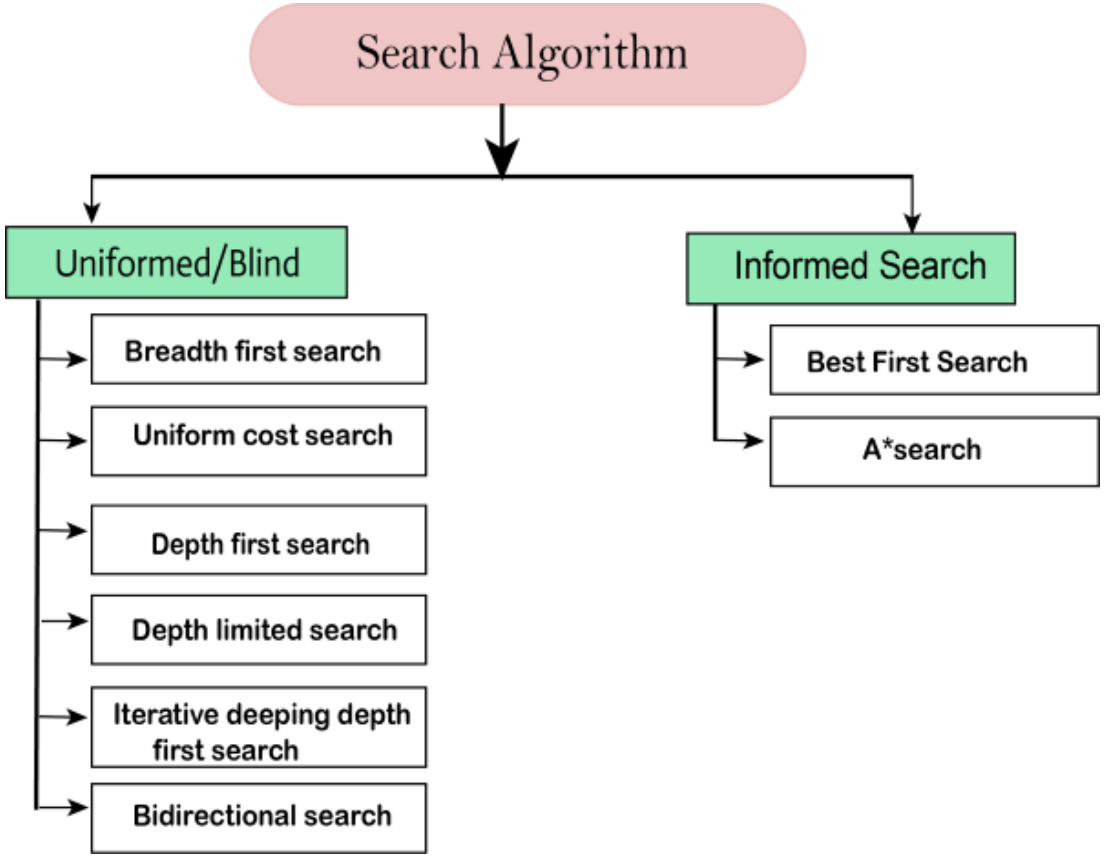
Search algorithm - infrastructure

- We need data structure to keep track of search tree
- For each node n of the tree:
  - $n.State$ - the state in the state space to which the node corresponds
  - $n.Parent$ - the node in the search tree that generated this node
  - $n.Action$ - the action that was applied to the parent to generate the node
  - $n.Path - Cost$ - $g(n)$ the cost of the path from the initial state to the node, indicated by parent pointers
- Nodes is different from state - we can have more nodes for the same state
- **Queue** - EMPTY?, POP and INSERT; FIFO queue, LIFO queue, priority queue

Measuring problem-solving performance

- Completeness - algorithm guarantees to return a solution if at least any solution exists for any random input
- Optimality - algorithm guarantees the best solution (lowest path cost) among all other solutions
- Time complexity - measures time for an algorithm to complete its task
- Space complexity - measures the maximum storage space required at any point during the search

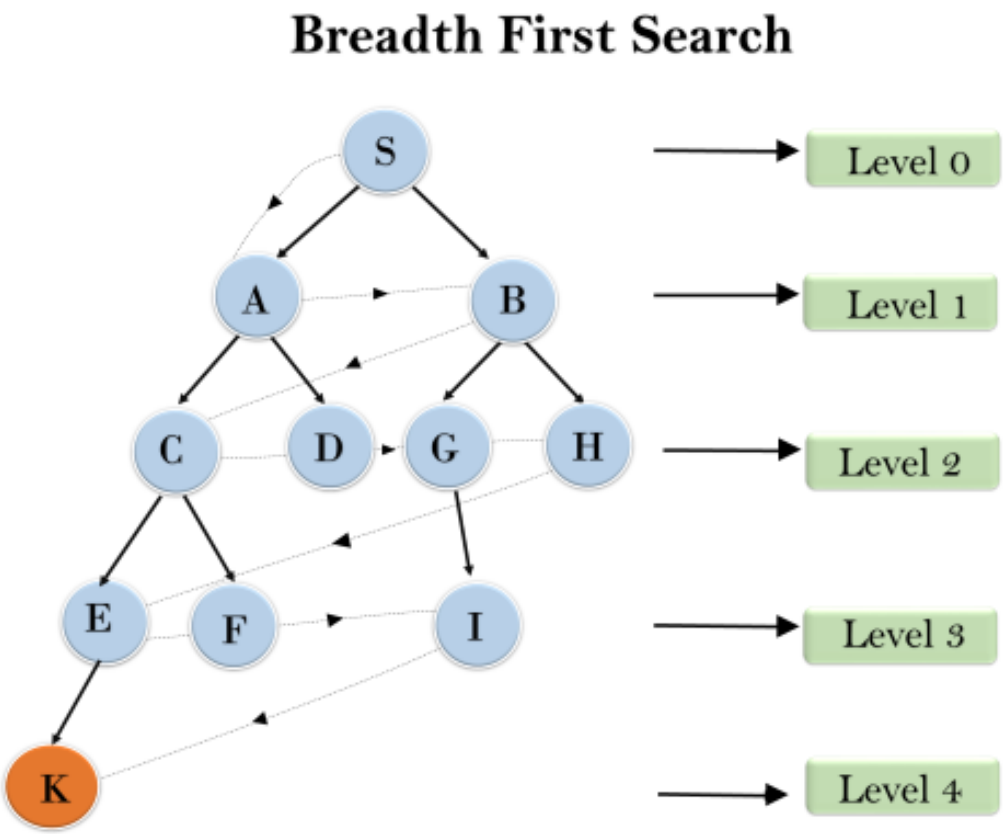
Types of search algorithms



Breadth-first search

- Starts searching from the root node of the tree and expands all successor node at the current level before moving to nodes of next level
- An example of a general-graph search algorithm; FIFO queue
- Advantages: provides a solution if exists; in the case with more than one solution provides the one with the least number of steps
- Disadvantages: requieres a lot of memory; needs a lot of time if the goal is far away from the initial state

Breadth-first search

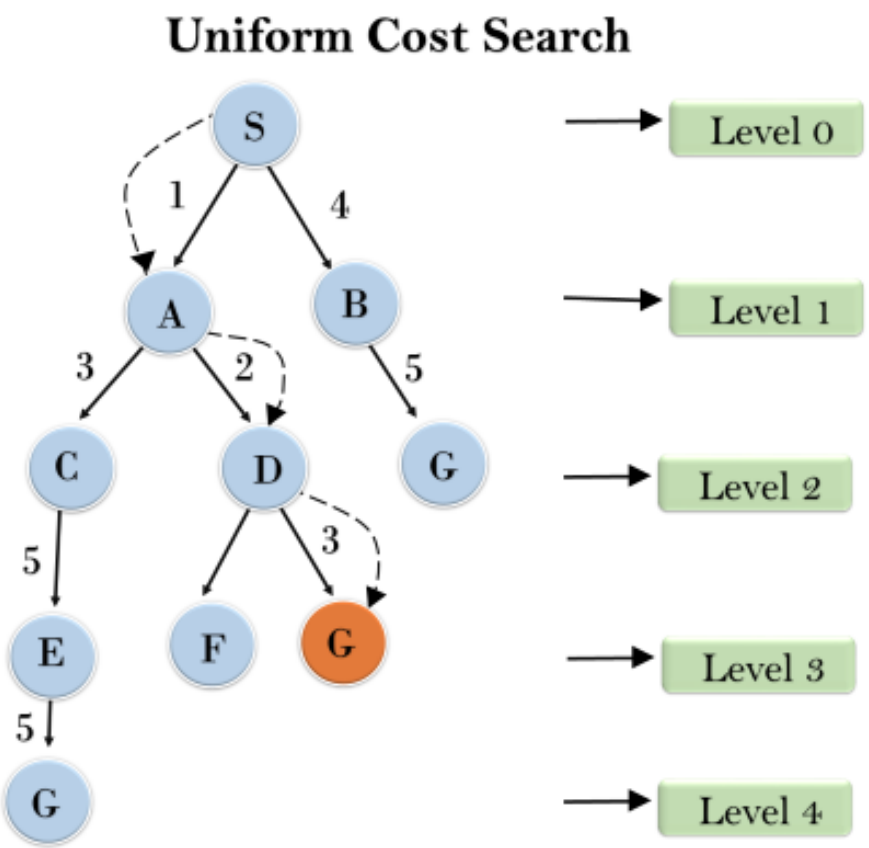


Time and space complexity $sO(b^d)$; complete if solution exists; optimal for non decreasing cost function

Uniform-cost search algorithm

- Used for weighted tree or graph; different costs for each edge
- Find path to goal state with lowest cumulative cost; priority queue
- Advantages: optimal because at every state the path with the least cost is chosen
- Disadvantages: does not care about the number of steps involve in searching and only concerned about path cost; it can be stackted in the infinite loop

Uniform-cost search algorithm

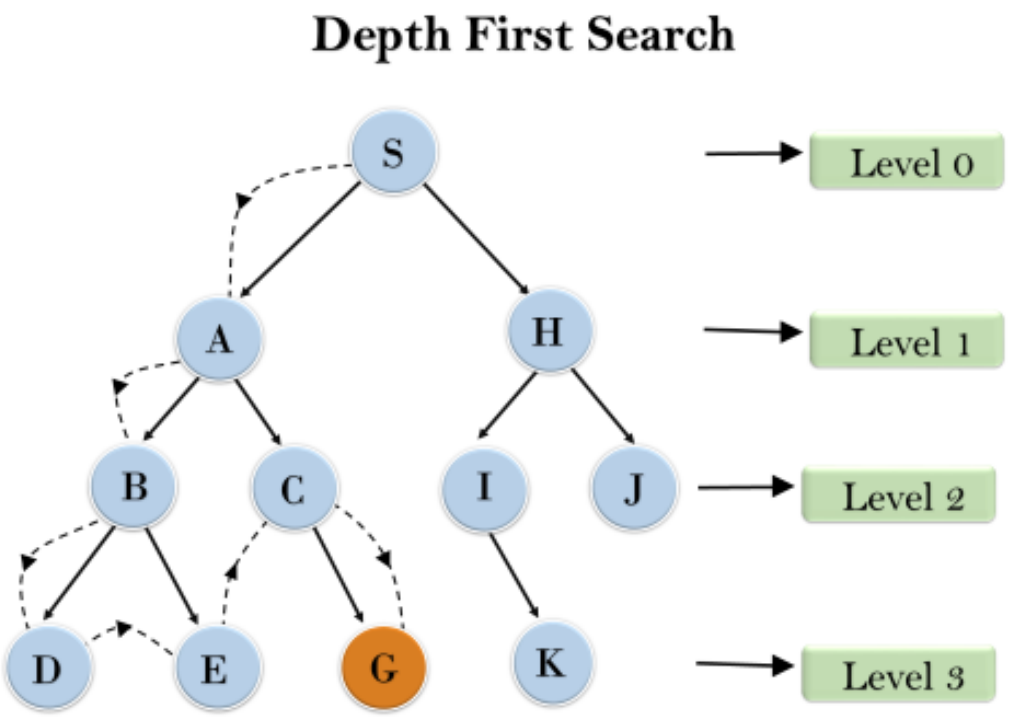


Time and space complexity: $O(n) = O(b^{1+[\frac{C^*}{\epsilon}]})$; complete; optimal - always choses path of the least cost

Depth-first search

- Starts from the root node and follows each path to its greatest depth node before moving to the next path
- LIFO queue; similar to breadth-first search (BFS)
- Advantages: less memory than BSF; less time to reach the goal node
- Disadvantages: the solution can re-occurre, no garantee for the solution; it can go to infinite loop

Depth-first search

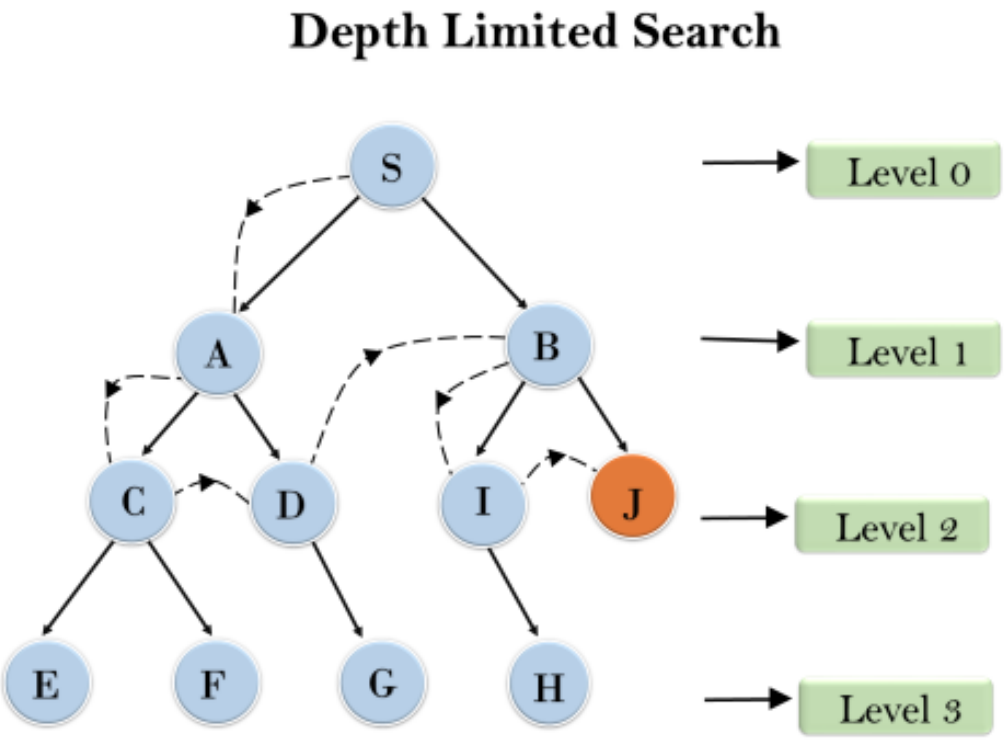


Time complexity: $O(n) = O(n^m)$; m - maximal depth of any node; for finite state space it is complete; non optimal; space complexity $O(b) = O(bm)$

Depth-limited search algorithm

- There is a set depth limit; even if there is a successor node after the depth limit it will disregard it
- Two types of faliure: standard failure value and cutoff failure value
- Advantages: memory efficient
- Disadvantages: incompletness; not optimal for problems with more than one solution

Depth-limited search algorithm

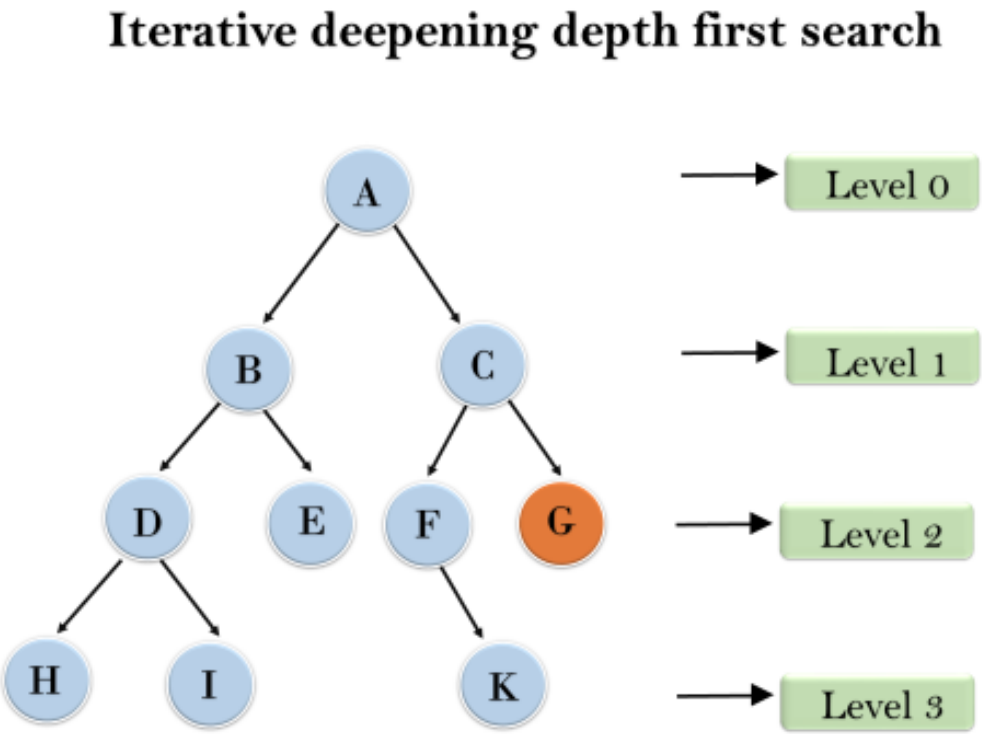


Time complexity: $O(b) = O(b^l)$; Space complexity: $O(b) = O(b * l)$; complete if solution is above depth limit; optimal

Iterative deepeningdepth-first search

- A combination of DFS and BFS; finds out the best depth limit and does it by gradually increasing the limit until a goal is found
- Performs depth-first search up to a certain "depth limit", and it keeps increasing the depth limit after each iteration until the goal node is found
- Advantages: fast search and memomry efficient
- Disadvantages: it repeats all the work of the previous phase

Iterative deepeningdepth-first search

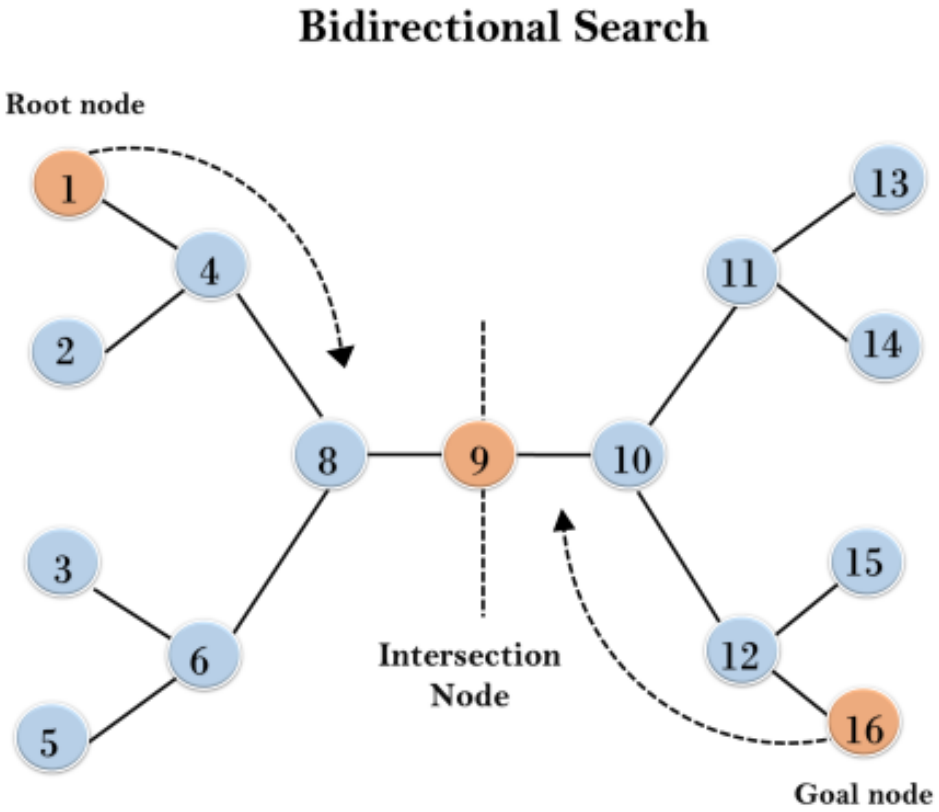


Time complexity: $O(b) = O(b^d)$; Space complexity: $O(b) = O(b * d)$; complete if the branching factor is complete; optimal for non decreasing cost function

Bidirectional search algorithm

- Two simultanious searches: forward-search from initial state and backward-search from the goal state
- Two smaller graphs; the search stops when they intersect
- It can use all described techniques
- Advantages: fast and memory efficient
- Disadvantages: complex implementation; we need to know the goal state in advance

Bidirectional search algorithm



Time and space complexity: $O(b) = O(b^d)$; complete if we use BFS for both searches; optimal

Informed search algorithms

- Informed search algorithm contains an array of knowledge such as how far we are from the goal, path cost, how to reach to goal node, etc.
- It helps agent to explore less to the search space and find more efficiently the goal node
- More suitable for large spaces; uses heuristics thus it is called heuristic search

Heuristic function

- Heuristic function is used in informed search to find the most promising path
- It produces estimation of how close is the agent from the goal based on its current state
- Heuristic method does not necessarily gives the best solution, but it guarantees to find good solution in the reasonable time
- $h(n)$ - it calculates the cost of an optimal path between the pair of states; it is always positive
- Admissibility of the heuristic function is given as $h(n) \leq h^*(n)$

Pure heuristic search

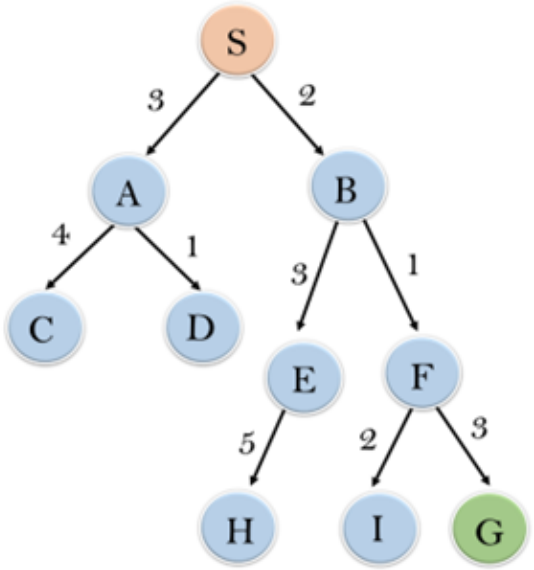
- The simplest form of heuristic search algorithms; expands nodes based on their heuristic value $h(n)$; two lists: OPEN and CLOSED list
- On each iteration, each node n with the lowest heuristic value is expanded and generates all its successors and n is placed to the closed list
- Algorithm stops when we reach the goal state
- Two types of algorithms: Greedy search algorithm and A^* search algorithm

Greedy search algorithm

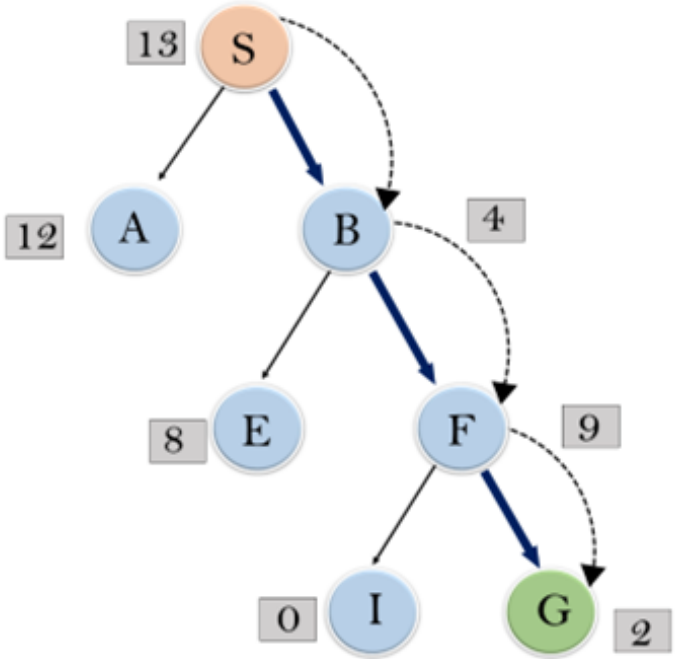
Always selects the path which appears best at that moment

- Step 1: place the starting node into the OPEN list
- Step 2: if the OPEN list is empty, Stop and return failure
- Step 3: remove the node n, from the OPEN list which has the lowest value of h(n), and places it in the CLOSED list
- Step 4: expand the node n, and generate the successors of node n
- Step 5: check each successor of node n, and find whether any node is a goal node or not; if any successor node is goal node, then return success and terminate the search, else proceed to Step 6
- Step 6: for each successor node, algorithm checks for evaluation function h(n), and then check if the node has been in either OPEN or CLOSED list; if the node has not been in any of the lists, then adds it to the OPEN list
- Step 7: Return to Step 2

Greedy search algorithm - example



node	H (n)
A	12
B	4
C	7
D	3
E	8
F	2
H	4
I	9
S	13
G	0



Time and space complexity: $O(bm)$, m - maximum depth of the search space; offten incomplete even in the finite space; not-optimal

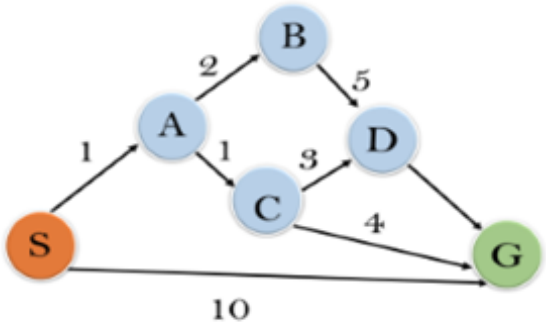
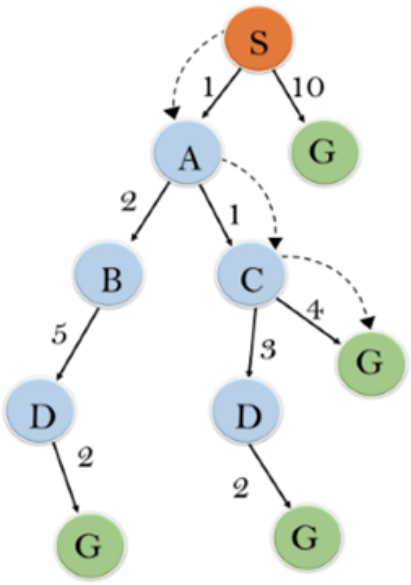
A* search algorithm

- Fitness number $f(n) = g(n) + h(n)$
- Advantages: the best algorithm than other search algorithms; optimal and complete; can solve very complex problems
- Disadvantages: does not always produce the shortest path as it mostly based on heuristics and approximation; has some complexity issues; memory demanding

A* search algorithm

- Step1: place the starting node in the OPEN list
- Step 2: check if the OPEN list is empty or not, if the list is empty then return failure and stops
- Step 3: select the node from the OPEN list which has the smallest value of evaluation function $(g + h)$, if node n is goal node then return success and stop, otherwise
- Step 4: expand node n and generate all of its successors, and put n into the closed list; for each successor n', check whether n' is already in the OPEN or CLOSED list, if not then compute evaluation function for n' and place into Open list
- Step 5: else if node n' is already in OPEN and CLOSED, then it should be attached to the back pointer which reflects the lowest g(n') value
- Step 6: Return to Step 2

A* search algorithm - example



State	h(n)
S	5
A	3
B	4
C	2
D	6
G	0